

NATHANIEL MCINTIRE

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EDUCATION

NORTH CAROLINA STATE UNIVERSITY

Expected Graduation: May 2027

Bachelor of Aerospace Engineering | Minor Astrophysics | Minor Mathematics

- Courses: Aerodynamics I & II, Aerospace Structures I & II, Intro. to Space Flight, Modern Physics, Astrophysics, Automatic Controls, Thermodynamics, Applied Machine Learning, Aerospace Vehicle Performance, MATLAB, SolidWorks

BRUNSWICK COMMUNITY COLLEGE

May 2024

Associates of Engineering | Associates of Science | Cum Laude Honors

ENGINEERING PROJECTS & DESIGN EXPERIENCE

HIGH-POWERED ROCKETRY CLUB

- Designed PCB layouts using KiCad and supported payload integration, ejection-system testing, and test preparation for high-power rocket systems. The PCB was designed solely because we could not purchase a small enough one for our rocket. Our PCB had four sensors, 2 accelerometers, a barometer, and a magnetometer.
- Fabricated composite components through carbon-fiber layups and assisted with evaluating payload and recovery-system performance.

AIRCRAFT WING AERODYNAMIC OPTIMIZATION

- Compared three airfoils in XFOIL across climb, cruise, and descent conditions and selected the NACA 23015 based on lift-to-drag performance, stability, and low-speed characteristics.
- Optimized a 50 m-span wing to AR 10 and a 0.5 taper ratio, calculating approximately 858 kN lift, 24.7 kN drag, and 138 kN-m wing-root torque at cruise.

CONVERGING-DIVERGING NOZZLE DESIGN AND FLOW ANALYSIS

- Designed and manufactured a converging-diverging nozzle in SolidWorks and developed an ANSYS CFD model to predict internal pressure and velocity behavior.
- Conducted wind-tunnel testing at NC State and processed experimental data in MATLAB to compare measured and computational flow characteristics.

IMPINGING-JET PRESSURE MEASUREMENT

- Measured the surface-pressure distribution produced by an impinging jet using binary pressure-sensitive paint.
- Developed a MATLAB image-processing workflow to convert experimental images into quantitative pressure-distribution data.

COMPOSITE LAMINATE STRESS ANALYSIS

- Evaluated ply-level stresses in composite laminates under multiple loading conditions and transformed stresses between material and global coordinate systems.
- Investigated how ply orientation and loading direction affected the structural response of laminated composites.

PENDULUM CONTROL-SYSTEM DESIGN

- Modeled a linearized pendulum system and designed a lead compensator using root-locus techniques to meet specified overshoot and settling-time requirements.
- Evaluated the compensated system response in MATLAB/Simulink and adjusted controller parameters to achieve the target transient behavior.

PLC AUTOMATION SYSTEM

- Programmed a PLC using ladder logic to control an automated pick-and-transfer sequence.
- Developed and tested the sequence logic governing actuator operation and object movement between locations.

FLIGHT-DELAY MACHINE-LEARNING MODEL

- Developed Gini-based CART, entropy-based CART, and random-forest models to predict commercial flight delays.
- Applied SMOTE and Tomek Links to address class imbalance and compared model performance using consistent evaluation criteria.

TECHNICAL SKILLS

MATLAB | Python | C++ | HTML | FEA | ANSYS | KiCAD | AutoCAD | SolidWorks | Machine Learning | Numerical Analysis | Orbital Mechanics | CFD | Simulink | PID Design | Root Locus Design | Propulsion Systems | Structural Analysis | Neural Networks | Lead Compensation | Ladder Logic | Project Management | XFOil | Materials | Quantum Mechanics | Semiconductor Design | Astrophysics | PLC Control

PROFESSIONAL EXPERIENCE

BONGIORNO & SON ITALIAN SPECIALTIES

March 2025 - Present

Supervisor

Raleigh, NC

- Supervise a 10-person team and oversee scheduling, inventory, cash handling, opening/closing procedures, and daily sales of \$3,000–\$6,500 in a fast-paced airport restaurant.
- Reduced labor costs by 10% while increasing average sales by 25% over three months, contributing to a sustained 5.0-star Google rating.

ACE HARDWARE

August 2024 - March 2025

Supervisor

Raleigh, NC

- Supervised 10 employees and coordinated inventory, cash handling, twice-weekly shipments, and approximately 10–20 customer deliveries per week.
- Assembled approximately 20 mechanical products per week, including grills, lawn equipment, and chainsaws, and operated a forklift to move loads up to 1,200 pounds.

LOWES FOODS

October 2020 - August 2024

Guest Services Manager

Wilmington, NC

- Managed a 40-person front-end operation, including scheduling, hiring, onboarding, cash-office operations, online orders, guest services, and facilities support.
- Safeguarded \$60,000–\$80,000 in daily transactions, improved promotional sales by 10%, and earned a 95% audit score with consistent 100% secret-shopper scores.